Infosys Springboard

AirFly Insights

Data Visualization & Analysis of Airline Operations

Infosys Springboard Internship Project

Presented By: Anand Kumar
Mentor: Priya
Email: anandsibbuchoudhary@gmail.com

PROJECT OBJECTIVE

Project Overview

📈 Problem Statement

The airline industry faces major challenges due to flight delays and cancellations. These delays affect airline performance, airport operations, and passenger satisfaction. Airlines need proper analysis to understand the reasons for delays and cancellations.

🔍 Objective

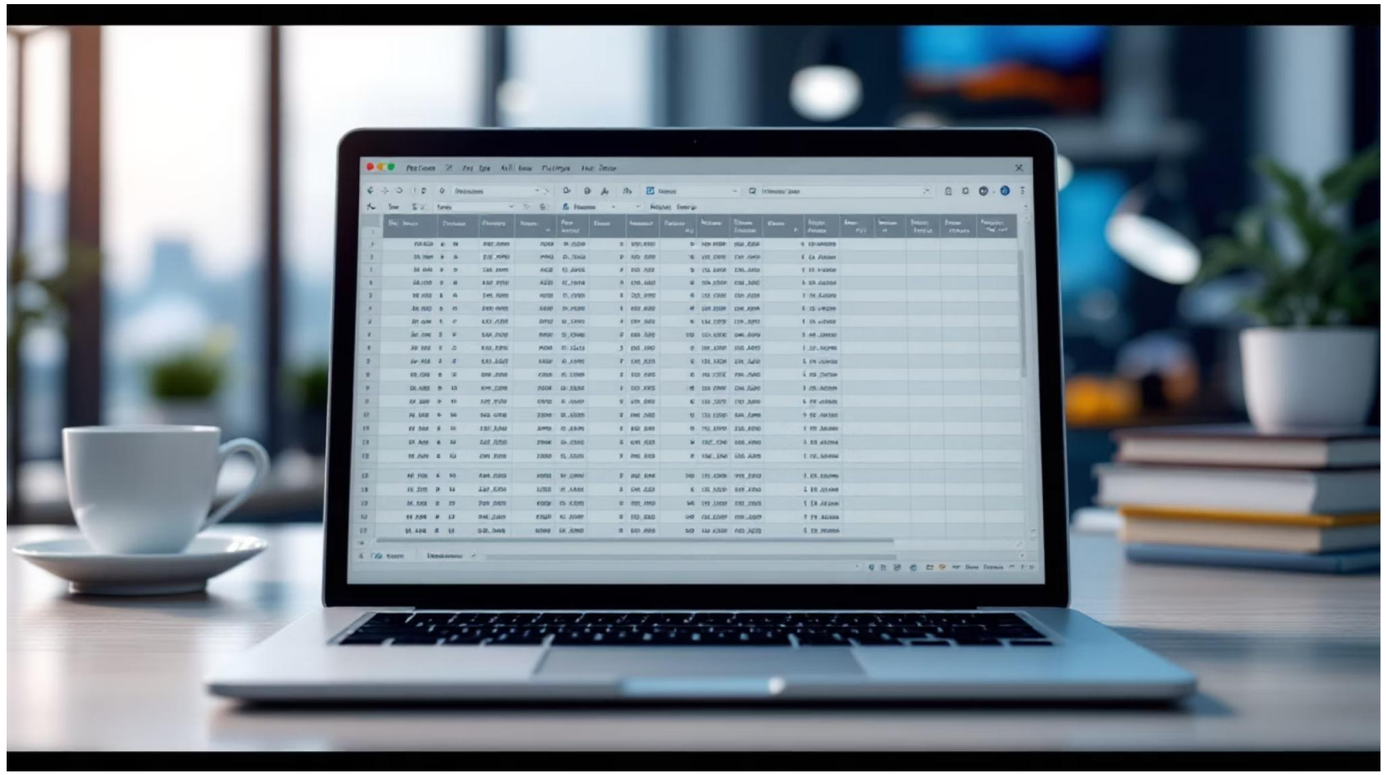
Analyze airline data to identify delay patterns, cancellation trends, and airport performance.

📊 Why This Project Is Important

- Helps airlines improve scheduling.
- Helps airports manage traffic.
- Helps reduce delays and cancellations.

DATASET

The Airline Flights Dataset



Sourced from **Kaggle**, this comprehensive dataset captures real-world airline operations across the United States, providing a rich foundation for exploratory and analytical work.

Dataset Contains:

- Airline information
- Origin and destination airports
- Departure and arrival delays
- Cancellation reasons
- Flight date and time
- Route information

Airlines & Routes

Carrier names, origin-destination pairs, and route frequency data.

Delays & Cancellations

Departure/arrival delays with categorized causes and cancellation codes.

Temporal Data

Flight dates, departure hours, and seasonal timestamps for trend analysis.

TECH STACK

Tools & Technologies Used

A modern Python-based stack was leveraged for data ingestion, transformation, analysis, and visualization throughout the project lifecycle.

Python

Core programming language for all data processing and analysis tasks.

Pandas & NumPy

Data manipulation, cleaning, and numerical computation at scale.

Matplotlib & Seaborn

Static visualizations for distributions, trends, and comparative analysis.

Plotly

Interactive charts enabling dynamic exploration of flight data.

Jupyter Notebook

Iterative development environment for analysis and documentation.

GitHub

Version control and collaborative code management for the project.

DATA PREPARATION

Data Cleaning & Preprocessing

Steps Taken

01

Handle Missing Values

Imputed or removed null entries in delay and cancellation fields.

02

Format Datetime Columns

Standardized date and time fields for consistent temporal analysis.

03

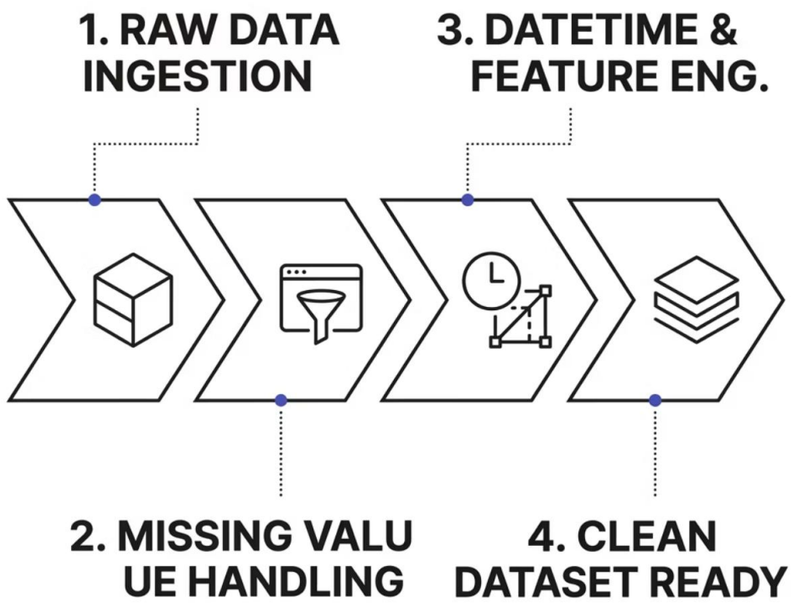
Feature Engineering

Derived new features: month, day of week, departure hour, and route.

04

Final Dataset Ready

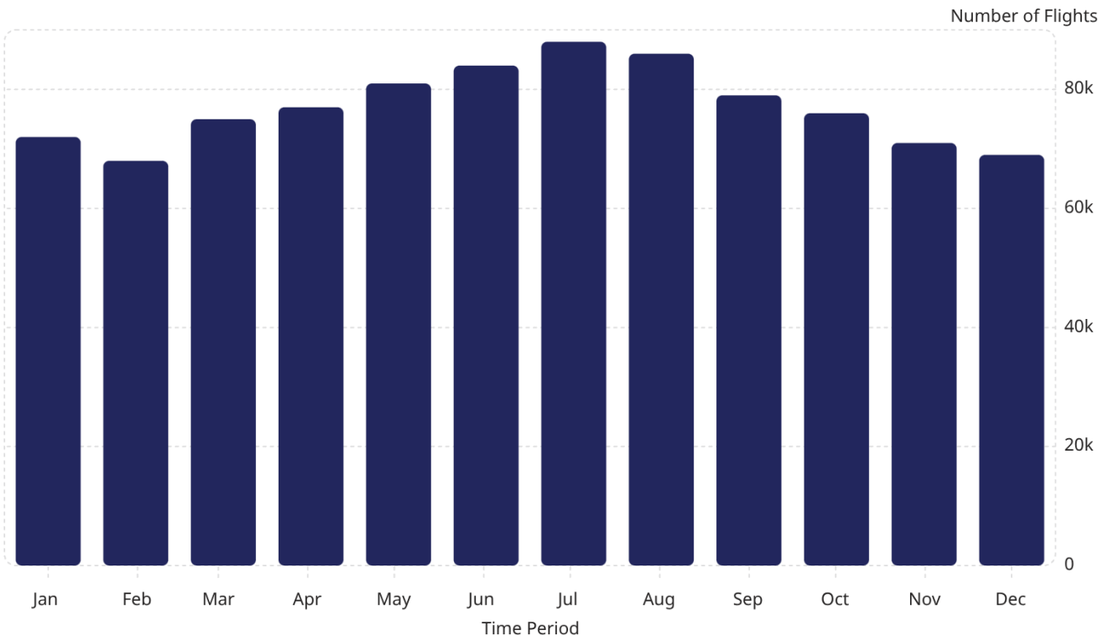
Cleaned data prepared and validated for downstream analysis.



EDA

Exploratory Data Analysis

Flight distribution was analyzed across multiple dimensions to reveal traffic patterns, peak travel periods, and operational rhythms in the dataset.

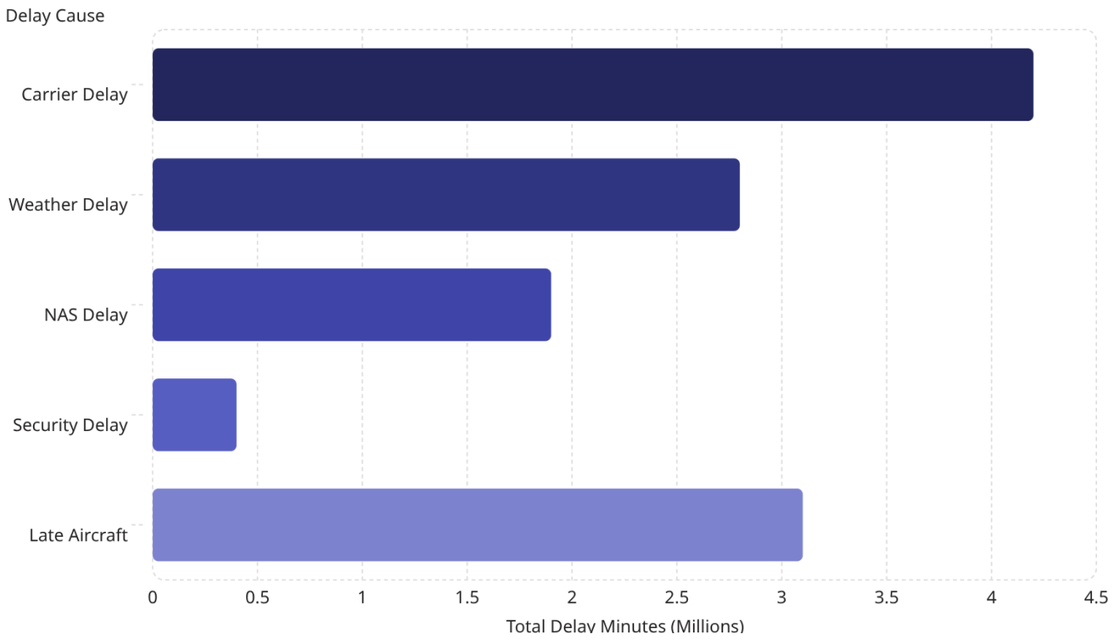


Key Observations

- **Peak Season:** Summer months (June–August) show the highest flight volumes.
- **Weekly Pattern:** Mid-week days see higher traffic compared to weekends.
- **Hourly Trend:** Departures peak between 6 AM–9 AM and 4 PM–7 PM.
- **Airline Share:** A few major carriers dominate overall flight frequency.

DELAY ANALYSIS

Understanding Flight Delays



What the Data Reveals

- **Carrier delays** account for the largest share of total delay minutes, pointing to operational inefficiencies.
- **Weather** is the second major contributor, especially impactful in winter months.
- **Evening flights** experience compounding delays due to cascading schedule disruptions.
- **NAS and Security delays** are relatively minor but consistent throughout the year.

Carrier + Weather delays together represent over **70%** of all delay minutes in the dataset.

ROUTE & AIRPORT ANALYSIS

Busiest Routes & Airport Congestion

🛬 Top Routes

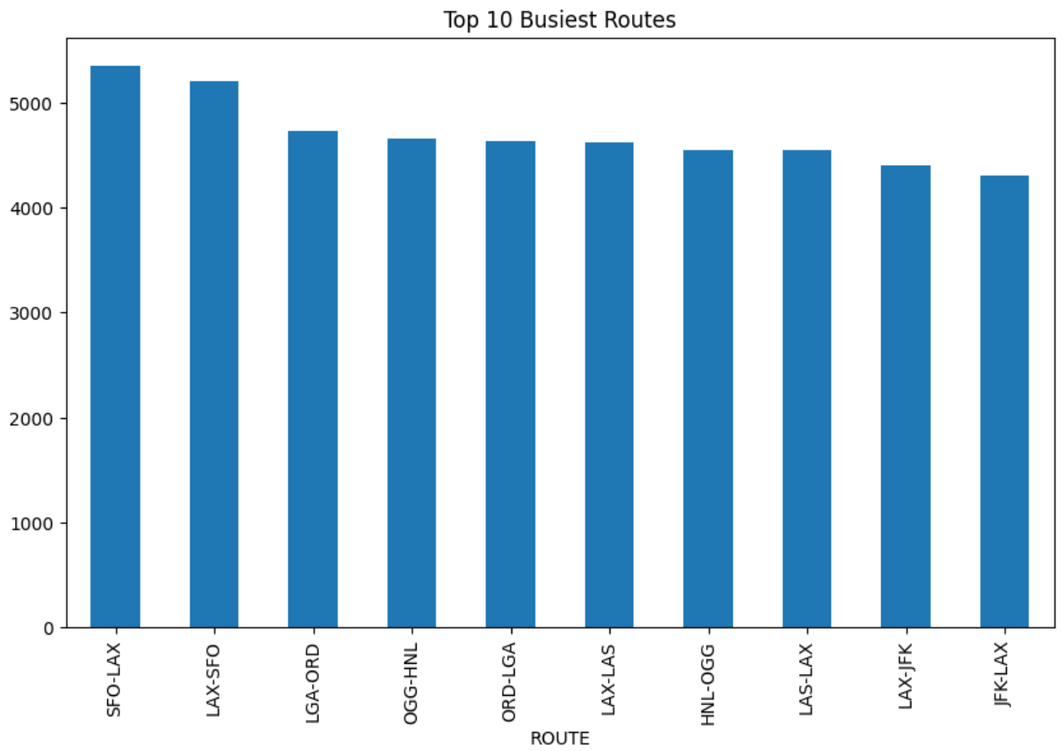
The top 10 busiest routes were identified, revealing high-frequency corridors concentrated between major metropolitan hubs.

⏱️ Airport Delays

Airport-level delay analysis highlighted which hubs experience the most congestion and contribute most to system-wide delays.

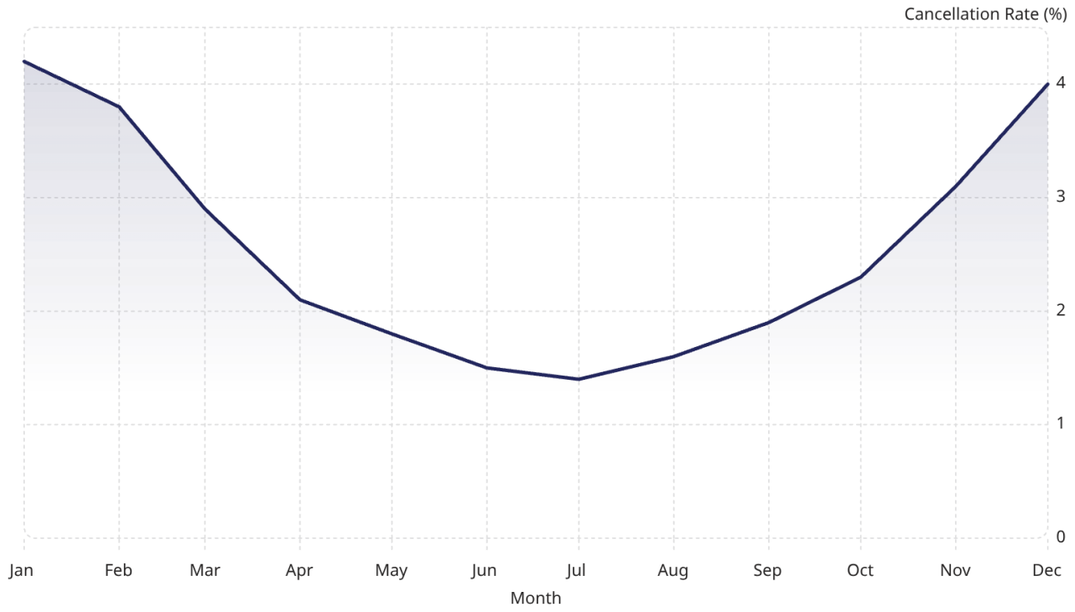
🔗 Route Performance

Comparing on-time performance across routes helps identify underperforming corridors that may need scheduling adjustments.



SEASONAL & CANCELLATION ANALYSIS

Cancellations & Seasonal Trends



Seasonal Insights

- **Winter spike:** Cancellation rates peak in January and December, driven primarily by adverse weather conditions.
- **Summer low:** July records the lowest cancellation rate, reflecting stable flying conditions.
- **Delay reasons:** Weather and carrier issues dominate cancellation codes across all seasons.
- **Recovery trend:** Cancellation rates decline steadily from February through summer.

Filters

AirFly Insights — Airline Operations Dashboard

Filters

Select Airline

All Airlines

Select Month

All Months

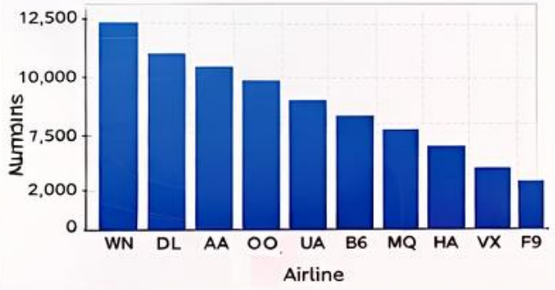
Delay Threshold (mins)

0 — 15 — 200

Apply Filters

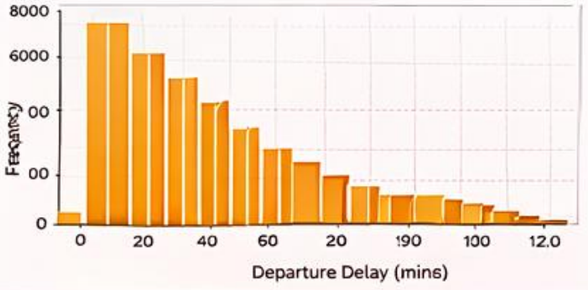
Total Flights	Delayed Flights	Cancellation %	Avg Departure Delay
58,206	13,742	1.92%	13.6 mins

Flight Distribution by Airline

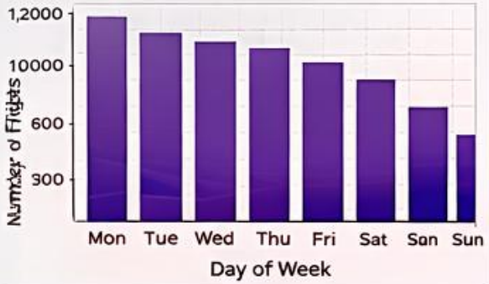


Flight Distribution by Month

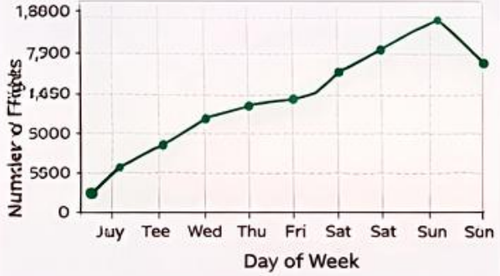
Departure Delay Distribution



Flights by Day of Week

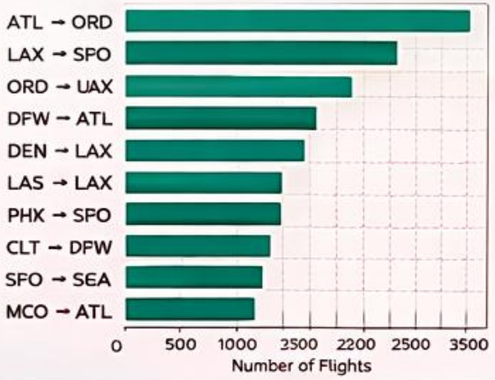


Flight Distribution by Airline



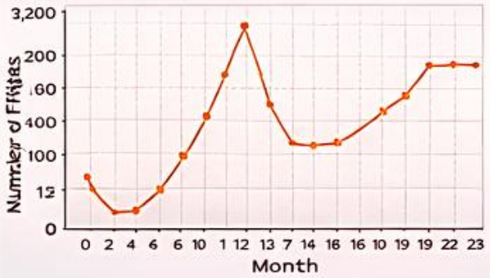
Flight Distribution by Month

Top 10 Busiest Routes

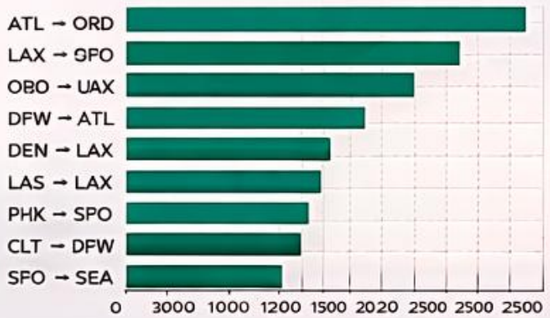


Flights by Day of Week

Flight Distribution by Departure Hour



Top 10 Busiest Routes



Visualization

CONCLUSION

Key Insights & Takeaways

Carrier & Weather Dominate

These two causes account for the majority of all flight delays across the dataset.

Evening Flights Are Riskier

Later departures face higher delay rates due to cumulative daily congestion.

Winter = More Cancellations

Cold-weather months consistently show elevated cancellation rates driven by weather events.

Visualization Drives Decisions

Clear visual analysis of airline data enables better operational and strategic decisions.

Thank You

This project was completed as part of the **Infosys Springboard Internship Program**. All analysis was conducted using publicly available data from Kaggle.